

KITABU QUEST

S5

Subsidiary Mathematics LFK HLP HGL

Mascot: mountain gorilla

Scene: students using applied mathematics for humanities and language combinations with a mountain gorilla mascot
CBC aligned draft learner book

Think • Explore • Achieve

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Title Page

KITABU QUEST S5 Subsidiary Mathematics LFK HLP HGL

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S5 learners study Subsidiary Mathematics LFK HLP HGL one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Learn through examples, activities, practice, and reflection.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Point, lines and geometric shapes in 2D
2. Functions and graphs
3. SEQUENCES

As you finish each topic, return to the success criteria and correct one answer before moving on.

Point, lines and geometric shapes in 2D: Lesson Opener

Learning focus: Point, lines and geometric shapes in 2D

Country link: a Rwanda school, market, farming, building, transport, or data problem for Point, lines and geometric shapes in 2D.

By the end of this topic, you should be able to:

- represent geometric shapes in 2D and calculate their area □ Learning objectives Content Learning Activities

Inquiry questions:

- How can point, lines and geometric shapes in 2d help you solve a real problem in Subsidiary Mathematics LFK HLP HGL?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Point, lines and geometric shapes in 2D: Learn

Point, lines and geometric shapes in 2D teaches properties of two-dimensional shapes.

Core idea:

- Shapes can be classified by sides, corners, lines of symmetry, and angles.
- A labelled sketch helps you compare shapes accurately.
- Use correct names such as triangle, rectangle, square, and circle.

Key words:

- Point, lines and geometric shapes in 2D
- Point
- lines
- geometric
- shapes
- method
- model
- unit

Point, lines and geometric shapes in 2D: Worked Example

Worked example for Point, lines and geometric shapes in 2D: Classify a shape with 4 equal sides and 4 right angles.

1. Four sides means it is a quadrilateral.
2. Equal sides and right angles match a square.
3. Check the name and properties.

Answer: The shape is a square.

Point, lines and geometric shapes in 2D: Vocabulary And Study Skill

Use these words and study moves before you practise Point, lines and geometric shapes in 2D. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Point, lines and geometric shapes in 2D
- Point
- lines
- geometric
- shapes
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Point, lines and geometric shapes in 2D without a clear example, method, or evidence.

Point, lines and geometric shapes in 2D: Guided Activity

Guided activity for Point, lines and geometric shapes in 2D:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Point, lines and geometric shapes in 2D: Practice And Correction

Practice for Point, lines and geometric shapes in 2D:

Main outcome: represent geometric shapes in 2D and calculate their area □ Learning objectives
Content Learning Activities.

1. Draw a triangle, rectangle, and square.
2. Write two properties for each shape.
3. Circle the shape that has four equal sides and four right angles.

Answer check:

Answer: the square has four equal sides and four right angles.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to point, lines and geometric shapes in 2d and solve it.

Point, lines and geometric shapes in 2D: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Point, lines and geometric shapes in 2D.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Point, lines and geometric shapes in 2D: Outcome Check

Outcome check for Point, lines and geometric shapes in 2D: Represent geometric shapes in 2D and calculate their area □ Learning objectives Content Learning Activities.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Functions and graphs: Lesson Opener

Learning focus: Functions and graphs

Country link: a Rwanda school, market, farming, building, transport, or data problem for Functions and graphs.

By the end of this topic, you should be able to:

- apply graphical representation of functions in solving economics and financial models

Inquiry questions:

- How can functions and graphs help you solve a real problem in Subsidiary Mathematics LFK HLP HGL?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Functions and graphs: Learn

Functions and graphs teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- Functions and graphs
- Functions
- graphs
- method
- model
- unit
- answer
- check

Functions and graphs: Worked Example

Worked example for Functions and graphs: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:

1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Functions and graphs: Vocabulary And Study Skill

Use these words and study moves before you practise Functions and graphs. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Functions and graphs
- Functions
- graphs
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Functions and graphs without a clear example, method, or evidence.

Functions and graphs: Guided Activity

Guided activity for Functions and graphs:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Functions and graphs: Practice And Correction

Practice for Functions and graphs:

Main outcome: apply graphical representation of functions in solving economics and financial models.

1. Solve one functions and graphs question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to functions and graphs and solve it.

Functions and graphs: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Functions and graphs.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Functions and graphs: Outcome Check

Outcome check for Functions and graphs: Apply graphical representation of functions in solving economics and financial models.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

SEQUENCES: Lesson Opener

Learning focus: SEQUENCES

Country link: a Rwanda school, market, farming, building, transport, or data problem for SEQUENCES.

By the end of this topic, you should be able to:

- apply arithmetic and geometric sequences to solve problems related to daily life situations

Inquiry questions:

- How can sequences help you solve a real problem in Subsidiary Mathematics LFK HLP HGL?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

SEQUENCES: Learn

SEQUENCES teaches how numbers change in a regular way.

Core idea:

- Look at how each term changes to the next term.
- A pattern can add, subtract, multiply, divide, or combine steps.
- Write the rule before extending the pattern.

Key words:

- SEQUENCES
- method
- model
- unit
- answer
- check

SEQUENCES: Worked Example

Worked example for SEQUENCES: Continue the pattern 6, 12, 18, 24, __, __.

1. Compare terms: $12 - 6 = 6$, $18 - 12 = 6$.
2. Rule: add 6 each time.
3. $24 + 6 = 30$, $30 + 6 = 36$.

Answer: 30, 36.

SEQUENCES: Vocabulary And Study Skill

Use these words and study moves before you practise SEQUENCES. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- SEQUENCES
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about SEQUENCES without a clear example, method, or evidence.

SEQUENCES: Guided Activity

Guided activity for SEQUENCES:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

SEQUENCES: Practice And Correction

Practice for SEQUENCES:

Main outcome: apply arithmetic and geometric sequences to solve problems related to daily life situations.

1. Continue the pattern 5, 10, 15, 20, __, __.
2. Write the rule for the pattern.
3. Create a similar pattern that increases by 4 each time.

Answer check:

Answers: 25, 30. Rule: add 5 each time.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to sequences and solve it.

SEQUENCES: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from SEQUENCES.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

SEQUENCES: Outcome Check

Outcome check for SEQUENCES: Apply arithmetic and geometric sequences to solve problems related to daily life situations.

1. Continue 7, 14, 21, 28, __, __.
2. Write the rule.
3. Create one new pattern with the same rule.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Point, lines and geometric shapes in 2D: Review Clinic 1

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from Point, lines and geometric shapes in 2D that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Functions and graphs: Review Clinic 2

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from Functions and graphs that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

SEQUENCES: Review Clinic 3

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from SEQUENCES that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Point, lines and geometric shapes in 2D: Review Clinic 4

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from Point, lines and geometric shapes in 2D that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Subsidiary Mathematics LFK HLP HGL.

- Point, lines and geometric shapes in 2D: Explain this term using an example from the book, your school, or your community.
- Functions and graphs: Explain this term using an example from the book, your school, or your community.
- SEQUENCES: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Subsidiary Mathematics LFK HLP HGL answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Subsidiary Mathematics LFK HLP HGL that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.